

Title: Identifying Conditions Favouring Multiplicative Heterogeneity Models in Network Meta-Analysis

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Configurations:

R version 4.5.1 (2025-06-13)

Platform: aarch64-apple-darwin20

Running under: macOS Tahoe 26.3.1

Matrix products: default

BLAS:

/System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks
/vecLib.framework/Versions/A/libBLAS.dylib

LAPACK:

/Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib
/libRlapack.dylib; LAPACK version 3.12.1

locale: [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Toronto tzcode source: internal

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] xtable_1.8-4 patchwork_1.3.2 ggpattern_1.3.1 gfmna_0.0.0.9000 [5] ggplot2_4.0.3 tibble_3.3.0
dplyr_1.1.4 netmeta_3.4-0

[9] meta_8.3-0 metadat_1.4-0 metabook_0.2-0

loaded via a namespace (and not attached):

[1] magic_1.6-1 generics_0.1.4 xml2_1.4.0 stringi_1.8.7
[5] lattice_0.22-7 lme4_1.1-37 hms_1.1.3 digest_0.6.37
[9] magrittr_2.0.4 grid_4.5.1 RColorBrewer_1.1-3 mvtnorm_1.3-3
[13] CompQuadForm_1.4.4 Matrix_1.7-3 purrr_1.1.0 scales_1.4.0
[17] numDeriv_2016.8-1.1 abind_1.4-8 reformulas_0.4.1 Rdpack_2.6.4
[21] cli_3.6.5 rlang_1.1.6 rbibutils_2.3 splines_4.5.1
[25] withr_3.0.2 tools_4.5.1 tzdb_0.5.0 nloptr_2.2.1
[29] minqa_1.2.8 metafor_4.8-0 colorspace_2.1-1 mathjaxr_1.8-0
[33] boot_1.3-31 vctrs_0.6.5 R6_2.6.1 lifecycle_1.0.4
[37] stringr_1.5.2 MASS_7.3-65 pkgconfig_2.0.3 pillar_1.11.1
[41] gtable_0.3.6 glue_1.8.0 Rcpp_1.1.0 tidyselect_1.2.1
[45] rstudioapi_0.17.1 farver_2.1.2 nlme_3.1-168 igraph_2.1.4
[49] readr_2.1.5 compiler_4.5.1 S7_0.2.0

Code execution process:

Before running any scripts, set the folder **Code Supplement** as the working directory.

The following scripts can be executed independently and do not require a specific order:

- **main_results.R**
Reproduces the histograms in Figures 1 and 10, as well as Tables 1 and 2.

- `casestudies1_4.R`
Reproduces the results for Case Studies 1–4, including Figures 2–8 and Tables 3–6.
- `casestudy_appen.R`
Reproduces the appendix case study illustrating the correct calculation of standard errors.

Source of data:

The `nmadb` package does not contain the datasets directly. Instead, the package retrieves data from an external online repository:

<https://cran.r-project.org/src/contrib/Archive/nmadb/?C=D;O=A>

Since the `nmadb` package is no longer actively maintained, we downloaded and locally saved all datasets used in this paper to ensure reproducibility. The script `save_data.R` provides the code used to download and store these datasets.

The datasets are stored in two formats:

- Individual `.rds` files in the folder `nmadb_twoarm_data/`
- A combined dataset file `nmadb_twoarm_data_all.rds`